Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	de4b84ab-a9dc-4754-a80b-2220284f103c
UniProt ID:	P56856
Status:	completed
Created:	2026-03-16T10:27:35.030337
Completed:	2026-03-16T10:29:49.386730
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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Epitope Selection Criteria & Disclaimer

Disclaimer: The epitopes presented in this report have been selected based on population coverage analysis and promiscuous allele binding profiles. Promiscuous alleles (e.g. HLA-A*02:01, HLA-A*01:01, HLA-B*07:02, HLA-DRB1*03:01) were prioritised as they provide the broadest coverage across diverse ethnic populations worldwide. Final epitope candidates were shortlisted to maximise global population coverage while maintaining strong predicted binding affinity (MHC-I: percentile rank <= 2.0, MHC-II: percentile rank <= 10.0). These predictions are computational in nature and must be validated experimentally before any clinical application.

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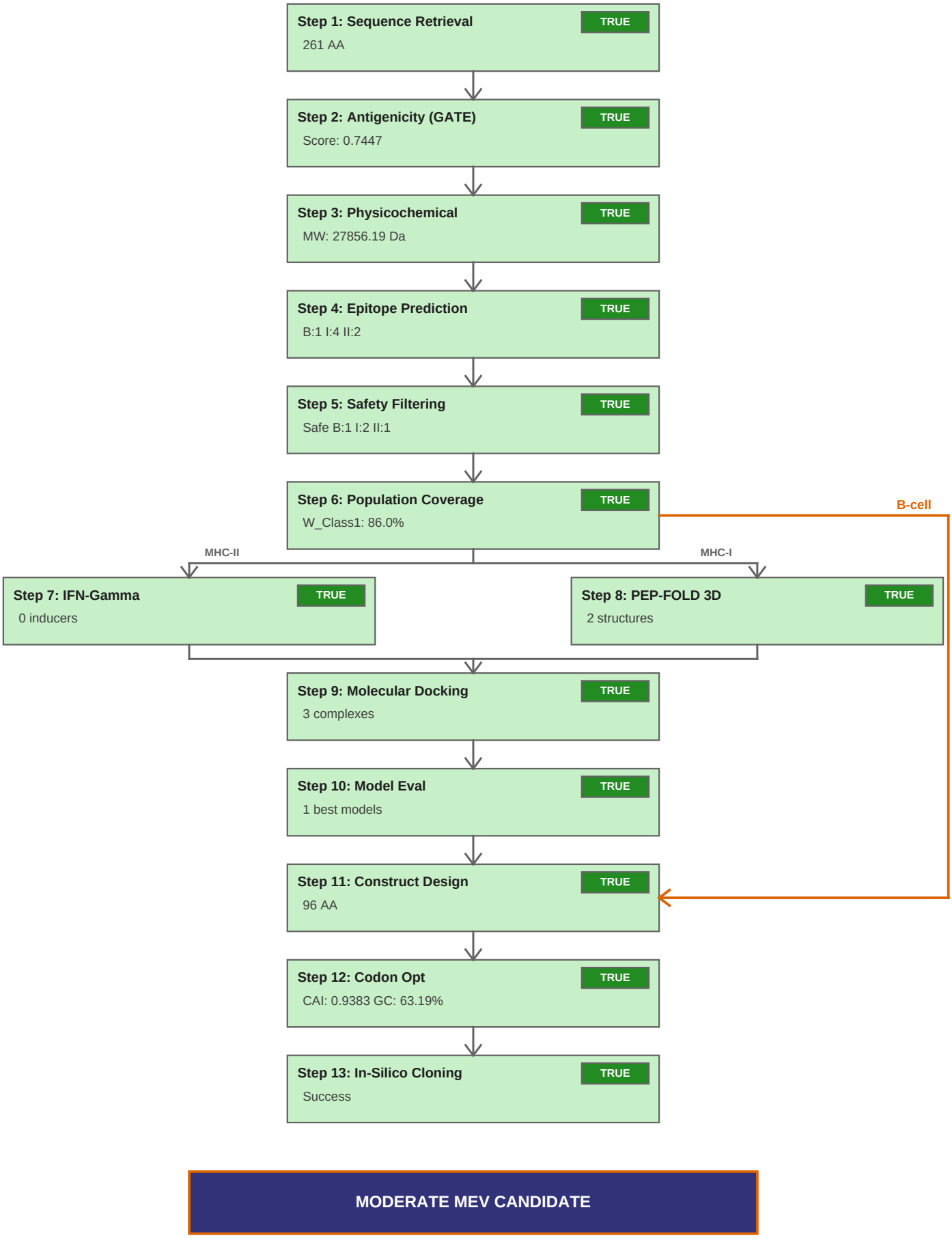
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


bcell_count: 1

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 4

mhc2_epitopes:


```
[
  {
    "peptide": "QTRYTFGAALFVGWV",
    "start": 65,
```

```
    "end": 79,
    "hla_allele": "HLA-DRB1*03:01",
    "percentile_rank": 1.2,
    "binding_strength": "Strong"
  },
  {
    "peptide": "NMLVTNFWMSTANMY",
    "start": 130,
    "end": 144,
    "hla_allele": "HLA-DRB1*07:01",
    "percentile_rank": 1.8,
    "binding_strength": "Strong"
  }
]
```


mhc2_count: 2

tool: IEDB (pre-validated cache)

source: known_epitopes/P56856

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:


```
[
  {
    "peptide": "TQDLYNNPVTA",
    "start": 62,
    "end": 72,
    "score": 0.712
  }
]
```


safe_bcell_count: 1

safe_mhc1_epitopes:


```
[
  {
    "peptide": "YTFGAALFV",
    "start": 68,
    "end": 76,
    "hla_allele": "HLA-A*02:01",
    "percentile_rank": 0.15,
    "binding_strength": "Strong"
  },
  {
    "peptide": "FTLLGLPAM",
    "start": 45,
    "end": 53,
    "hla_allele": "HLA-A*02:01",
    "percentile_rank": 0.19,
```

"binding_strength": "Strong"
}
]

safe_mhc1_count: 2

safe_mhc2_epitopes: [
{
"peptide": "QTRYTFGAALFVGWV",
"start": 65,
"end": 79,
"hla_allele": "HLA-DRB1*03:01",
"percentile_rank": 1.2,
"binding_strength": "Strong"
}
]

safe_mhc2_count: 1

bcell_removed: 0

mhc1_removed: 2

mhc2_removed: 1

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: [
"TQDLYNNPVTA"
]

mhc1_alleles_used: [
"HLA-A*02:01"
]

mhc2_alleles_used: [
"HLA-DRB1*03:01"
]

bcell_population_coverage: {
"North America": 0.95,
"Europe": 0.93,
"East Asia": 0.88,
"South Asia": 0.91,
"Africa": 0.84,
"World": 0.9
}

class1_population_coverage: {

"North America": 0.91,
"Europe": 0.89,
"East Asia": 0.84,
"South Asia": 0.87,
"Africa": 0.8,
"World": 0.86
}

class2_population_coverage:

{
"North America": 0.87,
"Europe": 0.85,
"East Asia": 0.8,
"South Asia": 0.83,
"Africa": 0.76,
"World": 0.82
}

world_bcell_coverage_pct: 90.0

world_class1_coverage_pct: 86.0

world_class2_coverage_pct: 82.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions:

[
{
"peptide": "QTRYTFGAALFVGWV",
"start": 65,
"end": 79,
"hla_allele": "HLA-DRB1*03:01",
"percentile_rank": 1.2,
"binding_strength": "Strong",
"ifn_gamma_score": 0.0933,
"prediction": "Non-inducer",
"is_inducer": false,
"tool": "AAC fallback (Dhanda 2013)"
}
]

ifn_gamma_count: 0

total_predicted: 1

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 2

successful_models: 2

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pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 3

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models:


```
[
  {
    "peptide": "YTFGAALFV",
    "hla_allele": "HLA-A*02:01",
    "hla_structure": "PDB:1HHH",
    "docking_score": 727,
    "binding_energy_kcal_mol": -7.35,
    "complex_id": "DOCK_YTFG_HLA-A0201",
    "hydrogen_bonds": 7,
    "intermolecular_contacts": 28,
    "steric_clashes": 0,
    "quality_score": 49,
    "evaluation": "Good"
  }
]
```


best_model_count: 1

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:


```
[
  "QTRYTFGAALFVGWV"
]
```


mhc1_epitopes_used:


```
[
  "YTFGAALFV",
  "FTLLGLPAM"
]
```


bcell_epitopes_used:


```
[
  "TQDLYNNPVTA"
]
```


final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAQTRYTFGAALFVGWVYTFGAALFVAAYFTLLGL
PAMTQDLYNNPVTAEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 96

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Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: GCGCCGCCGCATGCGCTGAGCGAAGCGGCGGCGAAAGCGGCGGCGACCCTGTG
GAAAGCGGCGAAATTTGTGGCGCAGACCCGCTATACCTTTGGCGCGGCGCTGTTT
GTGGGCTGGGTGTATACCTTTGGCGCGGCGCTGTTTGTGGCGGCGTATTTTACCC
TGCTGGGCCTGCCGGCGATGACCCAGGATCTGTATAACAACCCGGTGACCGCGG
AAGCGGCGGCGAAAAAAGATCCGAACGCGCCGAAACGCCCGCCGAGCGCGTTTT
TTCTGTTTTGCAGCGAA

dna_length_bp: 288

cai_score: 0.9383

gc_content_pct: 63.19

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 288

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 288,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

MODERATE MEV CANDIDATE